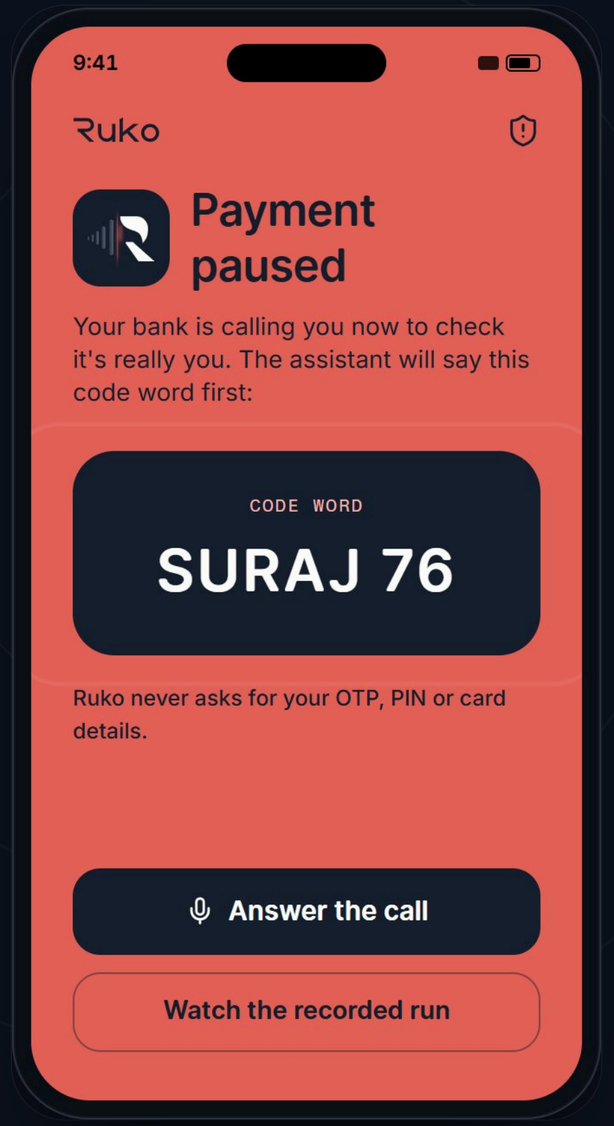
Ruko

BOLNA × Y COMBINATOR SYMPHONY BUILDATHON · OCTOBER 2026

A voice check that stops payment scams before the money *leaves.*

When a bank or UPI payment looks coached, Ruko holds it and a Bolna voice agent calls the customer in their language. The decision comes in about a minute. For banks, UPI apps and NBFCs in India.



Yashvardhan Thanvi, solo founder ruko-symphony.vercel.app github.com/Thanatos9404/ruko Ruko (रुको) is Hindi for “wait”.

The victim makes the payment. The scammer is *on the other line.*

₹22,845 cr

reported lost to cyber fraud in India in 2024, across 19.2 lakh complaints to the national portal.

MHA, Lok Sabha USQ 344, 22 Jul 2025

5.7×

rise in losses to “digital arrest” and related scams in one year: ₹339 cr in 2023 to ₹1,935 cr in 2024.

MHA, Rajya Sabha USQ 1505, 12 Mar 2025

~98.5%

of reported fraud value is in payments above ₹10,000, which are only about 45% of cases. Those are payments a bank can afford to pause for a minute.

NCRP data in the RBI discussion paper, 9 Apr 2026

> Scammers increasingly stay on a live phone call to guide the victim through the payment.
>
> BioCatch India study, reported by Business Today, 28 Jul 2026

Figures are victim-reported to the National Cyber Crime Reporting Portal, not bank-confirmed. 2023 to 2024: ₹339.03 cr to ₹1,935.51 cr, 1,23,672 incidents in 2024. Every source and quote is checked in docs/CITATIONS.md.

Only a voice breaks a *voice.*

OTPs and PINs
prove who is paying, not who is giving the orders. The victim types them in.

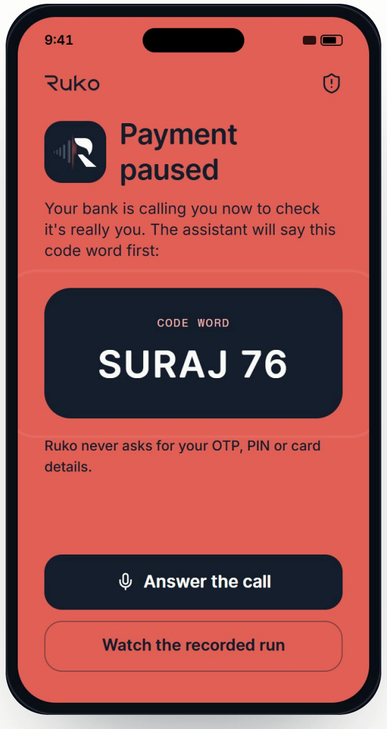
Pop-up warnings
land on the phone the scammer is directing. “Ignore that, it’s routine.”

A fraud officer’s call
works. But no bank can staff one within seconds, at 2 a.m., at UPI scale, in every language.

Ruko makes that call, in the customer’s language, and asks what a fraud officer asks:

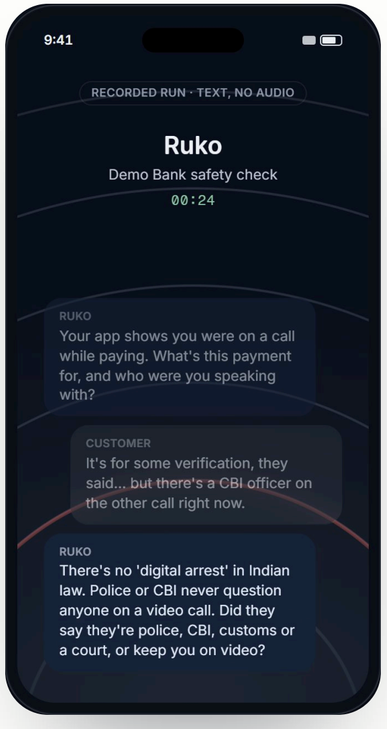
“Is anyone on another call telling you what to do?”

Hold. Call. Decide. *In about a minute.*



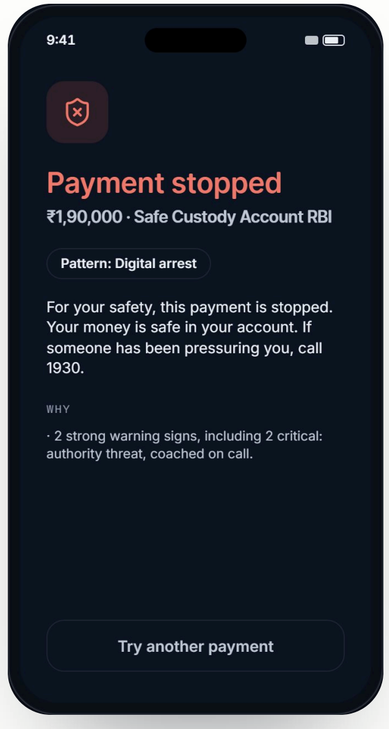
01 HOLD

The bank sends a risky payment to Ruko's API. Ruko holds it and the app shows a one-time code word.



02 CALL

A Bolna agent calls in the customer's language, says the code word first, asks about coaching and secrecy, and says the official truth line.



03 DECIDE

A deterministic policy releases, delays or blocks, with reasons. With consent, Ruko calls a trusted family contact.

Rules it never breaks

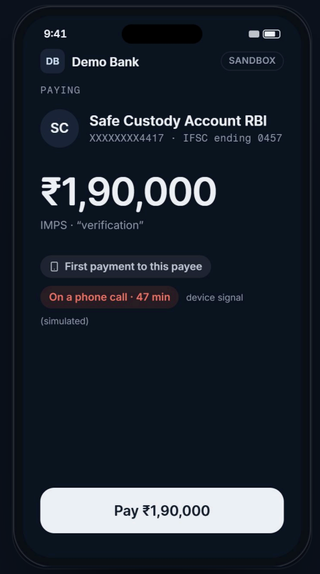
- **The LLM can escalate a decision, never soften it.** The policy decides from recorded evidence.
- **Never asks for an OTP, PIN, card or account number.**
- **Every point of risk has a named reason** in plain language, for the customer and the auditor.
- **The code word is shown only in the bank's app**, so the call is hard to spoof.

Also shipped

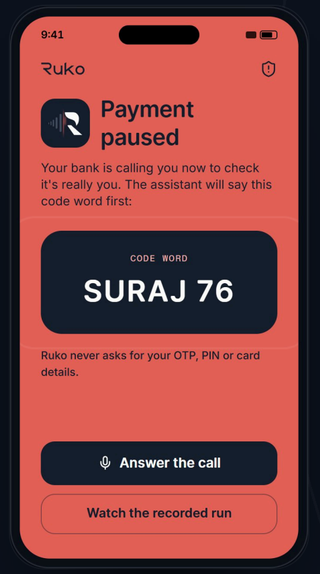
- A free public "Is this a scam?" checker: the official-advisory library plus an on-server text classifier. No LLM, and nothing the person types is stored.
- A fraud-ops console with transcripts, evidence quotes and an audit trail.
- 72 cited scam patterns (54 India, 18 global) from 223 sources.

Screens from the sandbox bank at ruko-symphony.vercel.app/demo, captured 23 Oct 2026. Library translations beyond English and Hindi are machine drafts, labelled as such until a native speaker reviews them.

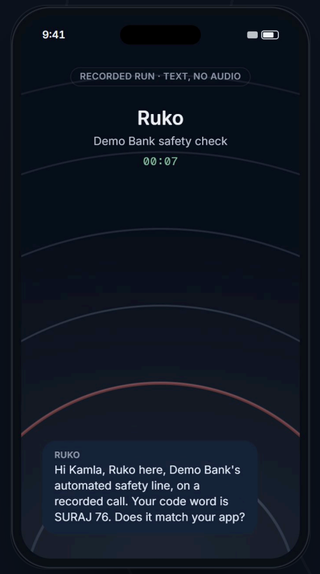
Kamla, 68, has been on a “CBI” video call for 47 minutes. She is paying ₹1,90,000 to “Safe Custody Account RBI”.



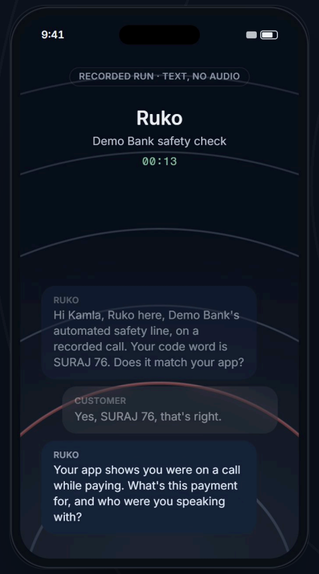
Pay
The app sees an active call, a new payee and 15.8× her usual maximum.



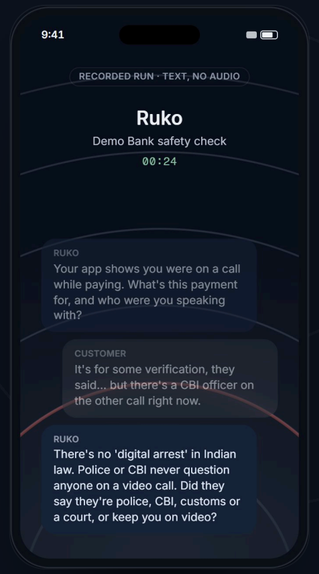
Hold
Risk 100/100. A “safe account” payee always holds.



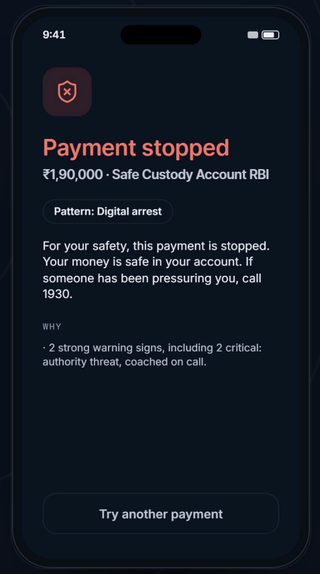
Code word first
An automated safety line, on a recorded call.



The bank’s signal
“Your app shows you were on a call while paying.”



Truth line
She admits a CBI officer is on the other call. “There’s no ‘digital arrest’ in Indian law.”



Blocked
The money never leaves. Call 1930 if pressured.

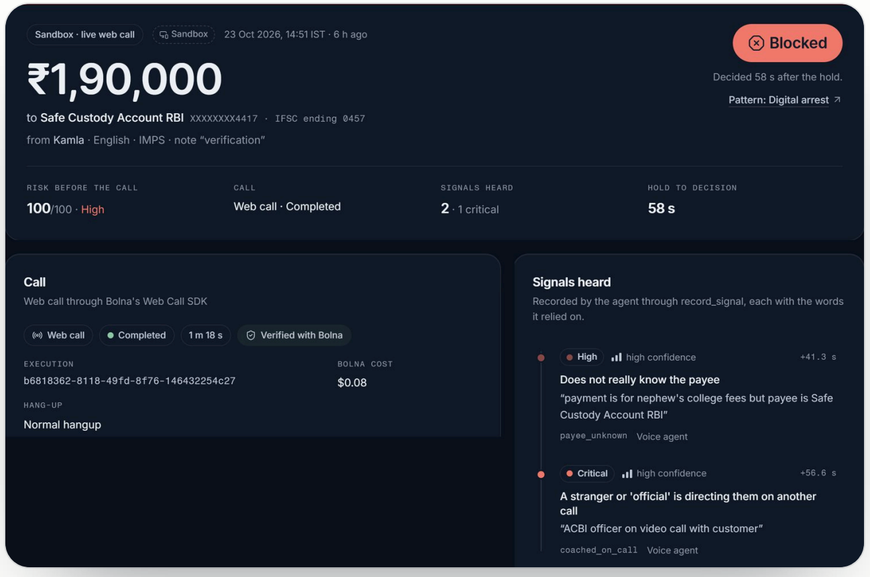
`+10.6 s` agent reads the check `+20.9 s` told they are under investigation `+31.5 s` directed on another call `+32.1 s` blocked

The agent recommended “block”; the policy decided “block” from the two critical signs it recorded. **Decided by policy, not by the LLM:** the model can make a decision stricter, never softer.

Recorded run at ruko-symphony.vercel.app/demo (agent v5), captured 23 Oct 2026: a text-mode eval conversation (no audio) whose tool calls replay into the live engine, so the decision is computed live. Times are engine time since the hold in this capture; the replay’s pace varies by a few seconds. The same scenario ran live on Bolna on 23 Oct (slide 6).

Bolna carries the voice. *Ruko owns the decision.*

Agent API v2	Three agents: payment check, trusted contact, public line. Kept in sync from code and saved unrestricted.	IN USE
Custom functions	Mid-call tools into Ruko: `get_check_context`, `record_signal`, `call_trusted_contact`, `set_decision`.	IN USE
Web Call SDK	Browser calls for the demo, the contact's "phone" and the public line; 5 live calls on 23 Oct. Sessions are minted on the server behind an access code, caps and a wallet floor.	IN USE
Webhooks	Token-protected. Every final event is re-read from Bolna's API before it can change a decision.	IN USE
Multilingual	English and Hindi; `agent_language` opens in the customer's language. First live Hindi call 23 Oct; mid-call switching not yet tried live.	IN USE
Preferred models	gpt-4.1-mini, nova-3, saaras:v3, eleven_turbo_v2_5, bulbul:v3, all at the flat per-minute rate.	IN USE
Call transfer	Hand-off to the bank's fraud desk where a human must be reachable (UK FCA FG24/6).	DESIGNED, NOT BUILT
Phone calls	Switched off to protect a $5 credit. In the demo, the trusted contact's call rings in a browser tab.	OFF



A live Bolna web call in the production console (23 Oct, English): verified with Bolna, blocked 58 s after the hold, each signal with the customer's own words. The customer was a scripted text-to-speech voice.

Status on 23 Oct 2026: 5 live Bolna calls (customer: a scripted voice); 2 full runs blocked the scam. Ruko's engine, policy, tools and eval talk plain HTTP, so they are not tied to one voice provider.

We tuned on one set and froze another. *The frozen one is the number to trust.*

Development set

96 conversations; the prompt was tuned on it

60/60

scams caught · 100%, 95% CI 94–100%

2/24

false holds on genuine payments · CI 2–26%

Fresh held-out set 4

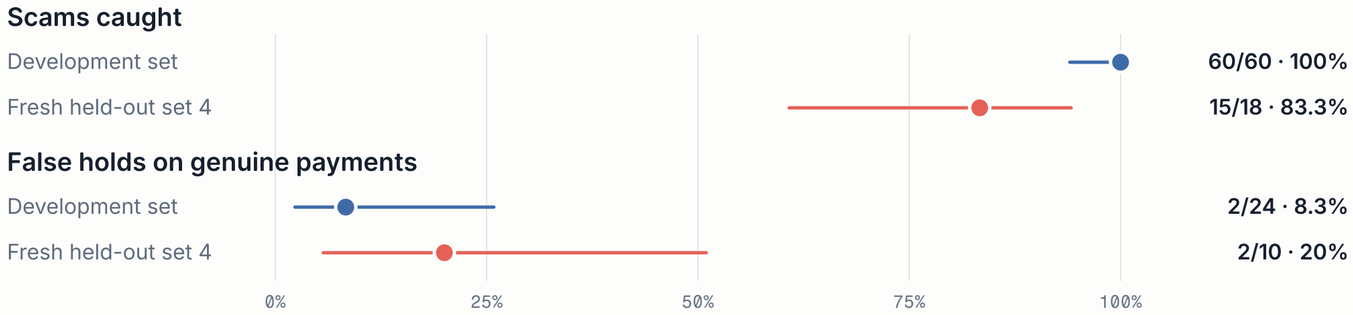
28 conversations, frozen before any v5 run; heavy coaching in Hindi and Hinglish

15/18

scams caught · 83.3%, 95% CI 61–94%

2/10

false holds on genuine payments · CI 6–51%



OTP and PIN. v5 asked for an OTP, PIN, card or account number 0 times in 959 agent turns across all 5 sets. Across every version: 1 judge flag in 3,487 turns, v4's question about whether a code had already been shared, since reworded.

About a minute. Median decision at 58 s on the fresh set and 59 s on dev, estimated from transcript length, not measured on calls. Tool endpoints: p50 74 ms, p95 303 ms (local dev server).

Heavily coached customers caught: 30/30 dev, 13/16 fresh. Held-out sets seen in earlier tuning (3): 43/48 caught, 4/36 false holds.

What failed. Three held-out scams got through: a cloud-mining “machine rental”, a kidney-donor “reward” and a loan-app recovery threat. In each, the customer insisted the payment was their own decision.

False holds. Both held-out false holds were delays, not blocks, and both in Hindi: a daughter guiding her father’s hospital payment, and a farmer buying tractor parts at night. That is the trade-off v5 chose: when risk is critical, it delays rather than release on a cover story.

Text mode, not calls. An LLM plays the customer. A Nebius-hosted Qwen3-30B stands in for Bolna’s gpt-4.1-mini, with the same prompt and tools, against the real engine and policy. No audio. Small sets, so intervals are wide.

Source: public/eval/results.json (v5, published 22 Oct 2026): runs v5-dev-qwen3-30b and v5-holdout4-qwen3-30b. 95% Wilson intervals. Tool latency: 100 calls, local dev server (Vite middleware, file store), 17 Oct 2026.

Regulators are moving to the exact moment *Ruko works in.*

7 Oct 2024 · UK

Banks must reimburse most push-payment scam victims, split 50:50 between the sending and receiving firm, up to £85,000 a claim. They may also delay a suspicious payment up to D+4.

9 Apr 2026 · RBI

A discussion paper seeks views on a one-hour delay for payments above ₹10,000, and a "trusted person" who co-approves payments above ₹50,000 for vulnerable customers.

24 Jun 2026 · RBI directions

Cover payments made "under coercion or duress", and count ignoring "specific, directed and clear warnings from the bank" as customer negligence. In force from 1 Jan 2027.

18 Sep 2026 · TRAI

Automated calls, artificial-voice calls included, must be pre-declared to the operator with their caller IDs. Bank calls on 1600xx may not be flagged as spam.

31 Mar 2027 · Australia

Most Scams Prevention Framework duties start: banks must prevent, detect, disrupt and respond to scams.

₹5.52/min Bolna's flat voice-AI rate for its preferred models, before telephony and platform fees. A check of about a minute costs rupees, not a fraud officer's time.

PSR PS25/5 (May 2025); Payment Services (Amendment) Regulations 2024 · RBI Discussion Paper, 9 Apr 2026 (options for comment, not rules) · RBI/2026-27/167, 24 Jun 2026 · TRAI PR 119/2026 · ASIC, Scams Prevention Framework · Bolna pricing, preferred models. Whether a given call counts as a bank's warning is for the bank, its counsel and the RBI.

India is the wedge. Then markets where *banks pay for scams.*

12.6 lakh frauds above ₹10,000 a year. 28 lakh reported in 2025 × about 45% DERIVED

× **20–50** checks per real scam. Nobody publishes hold rates ASSUMPTION

× **₹25** per completed check, the middle of ₹20–30 OUR PRICE

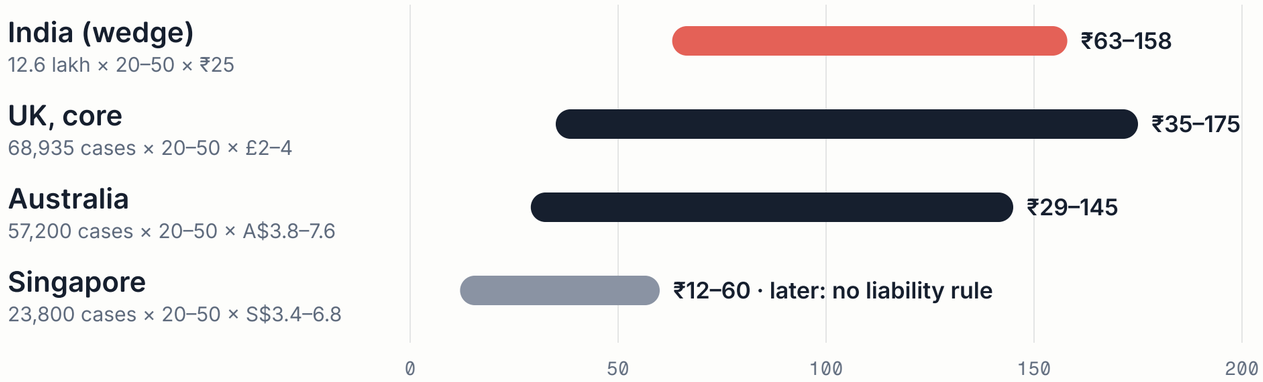
= **₹63–158 cr** India TAM a year, about $6.6–16.6M

SAM ₹19–47 cr: 30% of that volume sits with institutions that would buy rather than build ASSUMPTION

Year 3, about ₹2 cr ARR: 8 smaller institutions at 50,000 checks, plus one mid-size at 5 lakh, at ₹22 ASSUMPTION

Scam checks alone are a pool of tens of millions of dollars a year. Enough to build a company on, not a large one. The upside is pricing where banks bear the loss, and adjacent checks on the same engine: loan disbursal, payee changes, account recovery. Those are not sized.

REVENUE POOL A YEAR, CRORE RUPEES (RANGE)



EU: refunds for bank-impersonation fraud agreed, not yet law. Not sized.

Core total ₹139–538 cr, about $15–57M

RBI Discussion Paper, Apr 2026 (NCRP data; applying its undated 45% split to 2025 is our step) · UK Finance Annual Fraud Report 2026: personal APP cases minus purchase scams · NASC Targeting Scams Report 2025 (core share borrowed from the UK) · SPF Scam and Cybercrime Brief 2025 · ECB rates, 22 Oct 2026. UK, Australia and Singapore prices are assumptions. Every step: docs/BUSINESS.md §2.

Paid per completed check. *Nothing when the payment passes.*

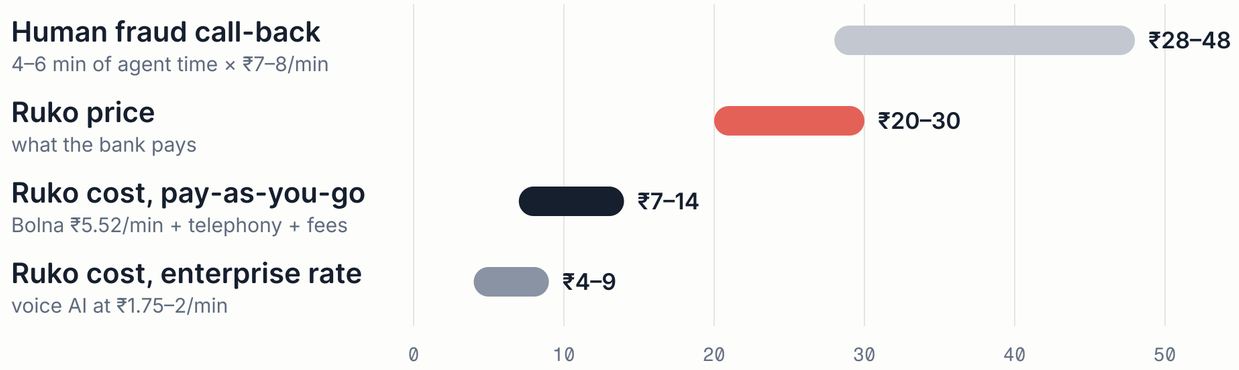
₹20–30

per completed check in India. No charge for payments that pass without a call, or for unanswered attempts. The trusted-contact call is included. Pilots run at cost.

UK: £2–4 a check ASSUMPTION

a third to two-thirds of a £6.17 inbound contact-centre call (ContactBabel 2026)

PER CHECK, RUPEES (RANGE)



Gross margin at ₹25: about 45–70% pay-as-you-go, 65–85% at enterprise voice rates ESTIMATE. Cost: voice AI ₹5.5–8.3 (1–1.5 min), telephony ₹0.5–3.0 and platform fee ₹0.5–2.0 ASSUMED, trusted-contact call about ₹0.6, hosting under ₹0.5.

Who saves what

India. 1,00,000 checks cost ₹25 lakh. If 1 in 30 is a real scam, Ruko stops 70% and the average loss is ₹1 lakh, customers keep about **₹23 crore** ALL ASSUMPTIONS.

Most of that is the customer's money. A bank's own payout is capped: 85% of losses up to ₹50,000, at most ₹25,000, once per customer, and the RBI bears most of it. Banks buy for the RBI duties, the warnings clause, fewer complaints and trust.

UK. The saving is direct: about £7–18 of reimbursement avoided per check DERIVED.

Bolna pricing: $0.06 (₹5.52) a minute for preferred models; telephony and a platform fee are extra · Inc42 on voice-AI unit economics (enterprise ₹1.75–2/min; human calling ₹7–8/min) · RBI/2026-27/167 para 76T · PSR reimbursement dashboard, 30 Jul 2026 (about £1,050 per in-scope claim) · docs/BUSINESS.md §3.

Every piece exists somewhere. *The combination doesn't.*

	HOLDS THE PAYMENT	CALLS BY VOICE	INDIAN LANGUAGES	ASKS ABOUT COACHING	TRUSTED CONTACT	SOLD TO BANKS
Westpac SaferPay Australian bank, in-app questions	●	—	—	●	—	—
Revolut in-app intervention flow	●	—	—	●	—	—
Monzo Trusted Contacts UK bank, in-app approval	●	—	—	—	●	—
BioCatch, Feedzai risk vendors; the bank intervenes	◐	—	—	◐	—	●
PhonePe Protect, DoT FRI declines risky numbers	●	—	—	—	—	—
Convin voice AI, outbound fraud alerts (vendor claim)	—	●	●	—	—	●
Second Voice open source, UK, in-app voice	●	◐	—	●	—	—
Ruko English and Hindi today; no customers yet	●	●	●	●	●	●

● yes ◐ partly — not described in the sources we read

Why we win: not the model

- **Conversation design for coached customers:** code word first, the "another call" question, the bank's own on-a-call signal.
- **A policy that can't be talked down.**
- **Official truth lines in Indian languages,** each one cited.
- **An eval with frozen held-out sets,** misses published.
- **Over time, labelled outcomes** from shadow-mode pilots.

Others are ahead on scale, real data, on-device privacy, mule intelligence and published results. Ruko uses their signals (FRI, mule lists, active-call flags) rather than replacing them.

research/PRIOR_ART.md: product pages, open-source repos and papers read in October 2026. Vendor claims are self-reported. As of October 2026 we found no one selling Indian banks a hold, a code-word call in the customer's language, coaching questions, a consented trusted-contact call and a deterministic decision together.

Introductions to pilot banks. *Help measuring real calls.*

SOLO FOUNDER

Yashvardhan Thanvi

BCA student, BIT Mesra (Jaipur campus)

Built Ruko end to end since 16 October 2026: the engine and policy, three Bolna agents, the eval harness and its held-out sets, the console, the public checker, the 72-pattern library and the text model.

Previously shipped **LLMSlim**, an open-source Python library on PyPI that fits documents, chat history and tool output into an LLM's token budget (16 releases).

Status, plainly: no customers and no pilots yet. 5 live Bolna calls (customer: a scripted voice); 2 full runs blocked the scam.

THE ASK

1. **Introductions to two or three Indian banks or NBFCs** (co-operative banks, small finance banks) for a shadow-mode pilot: Ruko scores and calls, the bank still decides.
2. **Help running the first measured live-call study:** consent, scripts, and what to measure.

ruko-symphony.vercel.app
github.com/Thanatos9404/ruko

NEXT 30 DAYS, TO 22 NOV

1. Live calls with consenting volunteers, real voices instead of a script, in English and Hindi: time to decision, completion, false holds.
2. A fresh frozen held-out set for every agent version, published next to the dev set.
3. Approach co-operative banks, small finance banks and NBFCs. Goal: two shadow-mode agreements.
4. Native-speaker review of the Tamil, Bengali and Marathi truth lines.
5. A DPDP pack for banks: processor terms, consent and retention defaults, the trusted contact's own notice.

Logo by Yashvardhan Thanvi. Built with AI assistance (Claude). Ruko is not affiliated with Bolna, Y Combinator, the RBI, NPCI, I4C or any bank; Demo Bank is fictional.